Cross-Dataset (orig_mod_timecorr.csv, samedata_r2_updated.txt) results analysis

1. Dataset Statistical Summary

Voltage Distribution

Statistic	Train Dataset	Test Dataset
Count	50,000	54,103
Mean (V)	40.8116	40.7149
Std (V)	0.3556	0.3167
Min (V)	37.8607	38.5789
25% (V)	40.5634	40.4915
50% Median (V)	40.9902	40.9165
75% (V)	41.0613	40.9165
Max (V)	41.1324	40.9873

Power Distribution

Statistic	Train Dataset	Test Dataset
Count	50,000	54,103
Mean (W)	399.41	402.67
Std (W)	127.06	139.41
Min (W)	193.39	201.91
25% (W)	337.31	327.65
50% Median (W)	384.72	381.36
75% (W)	447.81	456.64
Max (W)	3737.95	3630.96

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2. Model Performance

Voltage Prediction Results

Model	MAE (V)	RMSE (V)	R^2
DNN	1.1828	1.4095	-18.812
DeepNN	1.4188	1.7630	-29.997
LSTM	0.7569	2.3181	-52.582

Power Prediction Results

Model	MAE (W)	RMSE (W)	R^2
DNN	17.11	25.06	0.9677
DeepNN	17.12	23.13	**0.9725**
LSTM	54.76	108.46	0.3947

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3. Voltage Prediction Analysis

The voltage prediction models exhibited poor generalization when trained on the **orig_mod_timecorr.csv** dataset and evaluated on the **samedata_r2.txt** dataset. The statistical summary shows that voltage values have a **very narrow distribution**, with standard deviations of only **0.356 V (train)** and **0.317 V (test)**. Because voltage varies within a very small range around approximately **40–41 V**, even small prediction errors significantly impact the coefficient of determination (R^2).

Although the mean voltage values between the two datasets are similar, there is a slight shift in distribution and variance. Such small shifts become critical for low-variance signals like voltage. As a result, all three neural network models produced negative R^2 values, indicating that the models failed to capture the underlying voltage behavior in the test dataset.

Among the models, the **LSTM achieved the lowest MAE (0.7569 V)** but still produced a highly negative R^2 due to the small variance of the voltage signal. The poor generalization suggests that voltage dynamics are influenced by complex factors such as battery regulation, power bus stabilization, and environmental conditions, which may differ between datasets. Consequently, models trained on one dataset struggle to accurately predict voltage behavior in another dataset.

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4. Power Prediction Analysis

In contrast to voltage prediction, the power prediction models demonstrated strong performance and generalization across datasets. The statistical analysis indicates that the power distributions in the training and testing datasets are highly similar. Both datasets have comparable means (**399.41 W vs 402.67 W**) and similar standard deviations (**127 W vs 139 W**), indicating that the operational power characteristics remain consistent across datasets.

Because power values exhibit significantly larger variance compared to voltage, neural networks can more effectively learn relationships between the input features and the target variable. The selected features—**current, heater_power, load_current, payload_status, eclipse_flag, and aoce_mode**—directly influence the power consumption of the satellite subsystem, enabling the models to capture the relationship effectively.

Among the evaluated models, the **Deep Neural Network (DeepNN) achieved the best performance**, with an R^2 score of **0.9725**, indicating that approximately **97% of the variance in power consumption** is explained by the model. The standard DNN also performed well with an R^2 of **0.9677**. However, the LSTM model performed significantly worse, suggesting that power consumption in this dataset is largely determined by instantaneous system states rather than temporal dependencies. Therefore, sequence-based models such as LSTM provide limited advantage for this regression task.

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5. Overall Observation

The experimental results highlight a clear difference between voltage and power prediction behavior. Voltage prediction suffers from **low signal variance and slight distribution shifts between datasets**, leading to poor model generalization. In contrast, power prediction benefits from **larger variability and stronger direct relationships with the input features**, allowing neural network models—particularly DeepNN—to achieve high predictive accuracy. These findings indicate that while deep learning models can effectively capture power system behavior, voltage prediction across different datasets remains a challenging problem due to the sensitivity of the voltage signal and underlying system dynamics.